

Definite integrals



1 Evaluate the following definite integrals:

a $\int_0^3 (4x + 5) dx$

c $\int_2^4 (6x + 5) dx$

e $\int_{-4}^5 (-8x + 3) dx$

g $\int_{-2}^0 9x^2 dx$

i $\int_{-2}^1 (x^2 + 4) dx$

k $\int_4^6 (9x^2 + 2x + 7) dx$

b $\int_{-2}^0 (10x + 4) dx$

d $\int_3^4 (-4x + 3) dx$

f $\int_{-10}^{-6} (-5x + 8) dx$

h $\int_{-1}^3 9x^2 dx$

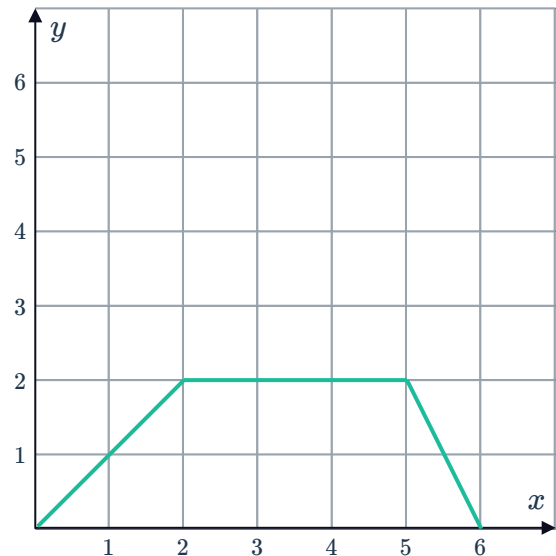
j $\int_{-1}^2 (9x^2 + 1) dx$

l $\int_{-4}^2 x(x - 4) dx$

Area under a curve

2 Consider the graph of $y = f(x)$.

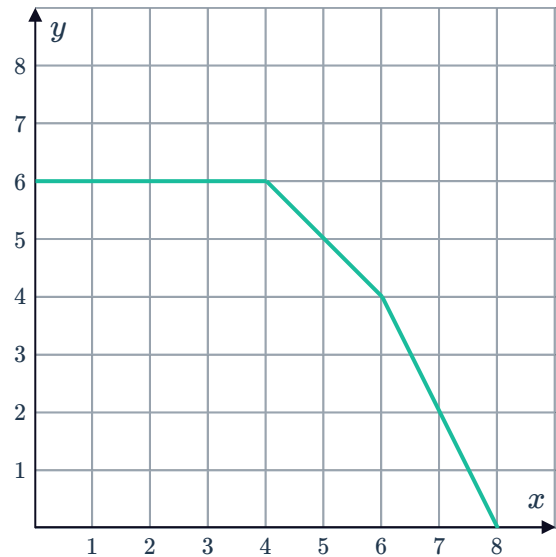
- a Find the value of $\int_0^2 f(x) dx$.
- b Find the value of $\int_2^5 f(x) dx$.
- c Find the value of $\int_5^6 f(x) dx$.
- d Hence state the area bounded by the function and the x -axis.
- e Write a single definite integral to represent the area bounded by the function and the x -axis.



3 Consider the graph of $y = f(x)$.



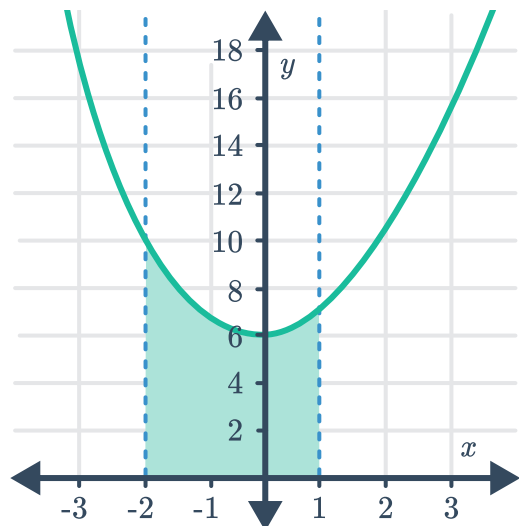
- a Find the value of $\int_0^4 f(x) dx$.
- b Find the value of $\int_4^6 f(x) dx$.
- c Find the value of $\int_6^8 f(x) dx$.
- d Hence state the area bounded by the function and the x -axis.
- e Write a single definite integral to represent the area bounded by the function and the x -axis.



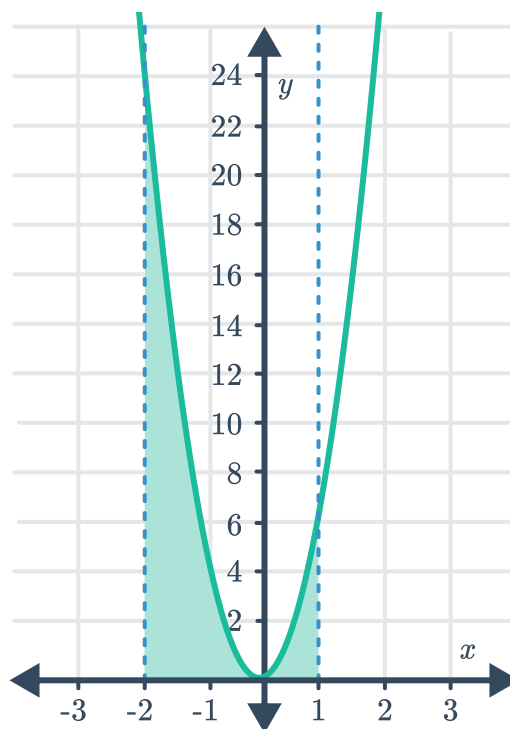
4 Consider the function $y = -(x + 2)(x + 8)$.

- a Find the x -intercepts of the function.
- b State the values of x for which the graph is above the x -axis.
- c Calculate $\int_{-8}^{-2} -(x + 2)(x + 8) dx$.
- d Hence find the area bounded by the curve, the x -axis and the bounds $x = -8$ and $x = -2$.

5 Consider the graph of the curve $y = x^2 + 6$. Find the exact area of the shaded region.



- 6 Consider the graph of the curve $y = 6x^2$.
Find the exact area of the shaded region.



- 7 Consider the given functions.

- i Sketch a graph of the function.
- ii Hence calculate the exact area bounded by the curve, $x = 1$, $x = 3$ and the x -axis.

a $y = 2x + 3$

b $y = -2x + 8$

c $y = 4x - x^2 - 3$